

Flexible Sensors for Shape Sensing and Tissue Analysis

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Abstract—This report presents a preliminary investigation into utilizing piezoresistive Carbon Nanotube (CNT) and Flexible Printed Circuit Board (FPCB) sensors for shape sensing and biological tissue analysis. To address the inherent non-linearities of CNT sensors, an initial calibration procedure was developed, and hysteresis compensation was discussed as a key requirement for future improvement. Furthermore, multi-layer tissue phantoms mimicking human skin were fabricated to evaluate the system's response during syringe needle insertion. While the preliminary tensile calibration and experimental setup design demonstrate the basic feasibility of this approach, further curvature tracking and pressure profiling experiments are still required for more complete validation. In parallel, a custom multi-channel FPCB impedance sensor was designed to differentiate between biological materials such as blood plasma, vessel walls, and clot material. Utilizing handmade single channel prototypes, preliminary testing demonstrated that by employing 2 ms pulsed DC voltages and a differential impedance (ΔZ) tracking algorithm, the system successfully compensates for the parallel conductivity of surrounding fluids. These initial findings provide crucial insights into the physical and algorithmic constraints of the setup, establishing a fundamental baseline and clearly defining the necessary hardware optimizations and advanced signal processing techniques required for future work.

Index Terms—Carbon nanotubes (CNTs), flexible electronics, shape sensing, hysteresis compensation, tissue phantoms, piezoresistive sensors.

I. INTRODUCTION

IN clinical procedures such as percutaneous punctures, physicians heavily rely on proximal haptic feedback—the mechanical resistance transmitted through the instrument shaft—to estimate distal tissue interactions. Transitioning from this subjective, experience-based “feel” to objective, direct localized sensing is a critical frontier in biomedical engineering. Traditional rigid sensors, however, suffer from a mechanical impedance mismatch when interfacing with soft biological structures. Consequently, highly conformable flexible sensors are required to seamlessly track dynamic surface deformations and accurately quantify the needle-tissue interface in real time. Alongside mechanical deformation tracking, evaluating the localized electrical impedance of tissues provides a crucial, complementary diagnostic modality. Differentiating between specific biological boundaries such as distinguishing blood plasma from vessel walls or clot formations requires conformable electrode arrays capable of operating accurately within highly conductive fluid environments.

Piezoresistive Carbon Nanotube (CNT) networks and Flexible Printed Circuit (FPCB) sensors offer exceptional strain sensitivity and flexibility, making them ideal candidates for dynamic shape sensing and localized pressure profiling. Despite their potential, deploying CNT-polymer composites introduces significant metrological challenges. The inherent viscoelasticity of the substrate and the microscopic friction within the nanotube network cause pronounced non-linearities and signal hysteresis. Addressing these dynamic distortions through rigorous calibration and algorithmic compensation is strictly necessary for reliable data acquisition.

This report presents a preliminary investigation into utilizing flexible CNT and FPCB sensors for deformation sensing and potential tissue puncture analysis. In the current stage, the work focuses on sensor mechanism analysis, electrical signal acquisition, preliminary tensile calibration, and the design of curvature and tissue phantom testing setups. Hysteresis compensation is discussed as an important requirement for future improvement rather than a fully implemented solution. Rather than presenting a finalized clinical tool, this study establishes a technical baseline and identifies the hardware and signal-processing challenges that must be addressed before reliable shape sensing and tissue-interface measurement can be achieved.

II. RELATED WORK

Flexible piezoresistive sensors have been widely studied for strain, pressure, and shape sensing because they can conform to curved or soft surfaces while converting mechanical deformation into electrical resistance changes. Carbon nanotube (CNT)-based sensors are especially promising due to their high conductivity, flexibility, and strong piezoresistive response [1], [2]. When CNT composites are stretched or bent, the conductive network between nanotubes changes, resulting in measurable resistance variation. However, CNT sensors often suffer from nonlinearity, signal drift, and hysteresis caused by polymer viscoelasticity and nanotube rearrangement, so calibration and compensation are necessary for reliable measurement.

Flexible printed circuit board (FPCB) sensors provide another useful platform for flexible resistance measurement. Compared with purely soft nanomaterial sensors, FPCB-based designs offer better electrical routing, easier integration with

readout circuits, and improved repeatability. Prior studies have used FPCB strain sensors for deformation monitoring in soft robotic structures and flexible devices [3]. These works support the feasibility of using flexible electronic structures for shape sensing and pressure measurement.

Artificial tissue phantoms are commonly used to evaluate biomedical devices under controlled conditions. Materials such as gelatin, agar, silicone, PDMS, and PVA can be tuned to mimic the mechanical properties of biological tissues [4]. Multi-layer phantoms are particularly useful for needle insertion studies because real skin contains different layers, such as the epidermis, dermis, and subcutaneous tissue. During needle insertion, force signals usually show gradual buildup, peak puncture force, and sudden force drops when the needle passes through tissue boundaries [5]. Therefore, combining flexible sensors with multi-layer tissue phantoms provides a practical method for studying tissue deformation, puncture behavior, and local pressure variation.

Overall, previous studies show that CNT and FPCB sensors are promising for flexible shape and pressure sensing, while tissue phantoms provide a controlled platform for biomedical testing. However, challenges such as CNT hysteresis, nonlinear calibration, unstable electrical contact, and indirect proximal force measurement still remain. This project builds on these prior works by evaluating flexible sensors for curvature sensing and tissue puncture analysis in a multi-layer phantom environment.

III. SENSORS DESIGN AND FABRICATION

A. Carbon Nanotube (CNT) Sensor

The primary flexible sensing element utilized in this project is based on a Carbon Nanotube (CNT) piezoresistive film. CNTs possess extraordinary electrical conductivity, high mechanical strength, and exceptional flexibility, making them ideal for integration with compliant substrates. The underlying sensing mechanism relies on the piezoresistive effect: when the flexible sensor undergoes mechanical deformation (such as bending or stretching), the dense, interconnected conductive network formed by the overlapping carbon nanotubes is dynamically disrupted. This structural reconfiguration alters the conductive pathways, leading to measurable variations in macro-scale electrical resistance.

1) *Sensor Architecture and Piezoresistive Mechanism:* In this project, pre-fabricated flexible Carbon Nanotube (CNT) piezoresistive sensors were utilized. These commercially available sensing elements consist of a highly uniform dispersion of multi-walled carbon nanotubes (MWCNTs) integrated with a thin, flexible insulating substrate, such as Polyimide (PI) or Polydimethylsiloxane (PDMS). Although the sensor fabrication was not performed in-house, understanding its underlying piezoresistive mechanism is crucial for accurate signal processing and calibration.

As the sensor substrate bends to a specific curvature, the outer surface undergoes tensile strain. In the microscopic view, this strain increases the mean distance between adjacent CNT bundles, breaking local electrical connections within the conductive percolation network, which consequently elevates

the total macroscopic resistance (R). Conversely, mechanical compression reduces the inter-tube distance, restoring these conductive pathways and decreasing the resistance. By continuously monitoring the relative resistance change ($\Delta R/R_0$), the exact state of dynamic deformation can be accurately quantified.

2) *Sensor Wiring and Signal Acquisition:* A major engineering challenge in nanomaterial-based flexible electronics is establishing robust, low-noise, and fatigue-resistant electrical connections between the microscopic CNT film and macroscopic copper wires. In our early iterations, standard bonding methods—including insulating tapes, heat-shrink tubing (applied to stabilize wire-sensor contact), and direct soldering—were evaluated. However, these methods consistently led to intermittent signal loss or catastrophic resistance fluctuations, primarily due to poor interfacial wetting, thermal degradation of the delicate CNT network, or insufficient contact pressure during flexing. To address this, a mechanical wrapping approach was deployed, where bare copper wires were tightly and securely coiled around the CNT sensor ends. This optimized configuration provided a stable, multi-point contact network that successfully eliminated signal drift and resisted mechanical fatigue, yielding highly reliable electrical interfaces.

Following the establishment of robust mechanical connections, the sensor was integrated into a simple yet highly effective voltage divider circuit for real-time signal acquisition. The CNT film, functioning as a variable piezoresistor (R_s), was connected in series with a precision fixed-value reference resistor (R_{ref}). A constant direct-current (DC) excitation voltage (V_{in}) was applied across the entire series circuit. By continuously monitoring the analog output voltage drop (V_{out}) across the sensing element (or the reference resistor), the instantaneous resistance of the CNT sensor could be dynamically back-calculated using the standard voltage divider principle:

$$R_s = R_{ref} \left(\frac{V_{out}}{V_{in} - V_{out}} \right)$$

This circuitry allowed the subtle resistance variations induced by mechanical deformation to be reliably converted into continuous voltage signals, which were subsequently fed into an analog-to-digital converter (ADC) for processing and data logging.

B. Custom Flexible PCB Design and Prototype Development

To transition toward a fully integrated and conformable diagnostic tool, a custom multi-channel Flexible Printed Circuit (FPCB) was designed. FPCBs are increasingly becoming the standard for accessing internal biofluids, as their highly conformable nature overcomes the tissue penetration limitations of traditional rigid sensors, enabling minimally invasive *in situ* impedance spectroscopy [10]. While the final multi-channel FPCB has not yet been fabricated, the underlying sensing methodology was rigorously evaluated using multiple handmade, single-channel prototypes of varying geometries.

The proposed FPCB design is specifically engineered for localized tissue analysis, distinguishing between blood plasma,

vessel walls, and blood clot material based on their distinct electrical impedances, a well-established modality for detecting blood coagulation and tissue boundaries [7], [8]. To map the spatial distribution of materials around the instrument, the finalized FPCB design features a multi-channel electrode array. This configuration allows the system to determine exactly which side of the probe is interacting with specific tissue types. During intended operation, the channels are driven using an alternating, time-multiplexed excitation strategy. By strictly alternating the active channels, the system design successfully minimizes parasitic cross currents and electrical interference between adjacent electrodes.

Furthermore, a continuous direct current (DC) would lead to adverse effects in a biological environment. Therefore, the sensor logic utilizes 2 ms pulsed DC voltages. This short pulse duration is critical for preventing localized Joule heating of the blood, mitigating fluid ionization, and eliminating the accumulation of charges (electrode polarization impedance) at the anode and cathode interfaces, as continuous DC is known to induce unwanted thermal and electrolytic effects in biological tissues [9].

IV. METHODOLOGY AND SIGNAL PROCESSING

A. CNT Sensor Calibration

To convert raw electrical signals into quantifiable physical parameters, a custom-built quasi-static tensile calibration procedure was employed. The CNT sensor was mounted onto a flat, rigid substrate, with one terminus securely affixed. Controlled tensile strain was applied by incrementally stretching the free end of the sensor, which was then temporarily anchored at specific displacement intervals. At each step, the absolute elongation was precisely measured using a precision vernier caliper, and the corresponding steady-state resistance was logged. Using these discrete datasets, a mathematical mapping between the relative change in resistance ($\Delta R/R_0$) and the applied tensile strain (ϵ) was established. A least-squares regression algorithm was then applied to extract the characteristic calibration coefficients, defining the sensor's gauge factor (sensitivity) and linear operating limits.

B. Shape Sensing and Curvature Measurement

To evaluate the sensor's shape-sensing capabilities, a geometric bending model was applied to correlate the calibrated tensile strain with the structural curvature. The experimental validation was conducted as follows:

- **Curvature Generation via Circular Loop Method:** To generate uniform and quantifiable bending deformations, flexible hoses of varying, predefined lengths (L) were formed into closed circular loops. This specific geometric configuration ensures a constant bending radius across the sensor, where $r = L/(2\pi)$. Consequently, the applied curvature ($\kappa = 1/r = 2\pi/L$) could be precisely controlled and stepped simply by adjusting the total length of the hose.
- **Strain-to-Curvature Analytical Mapping:** A fundamental kinematic model was employed to translate the

linear stretching response into bending curvature. During flexion, the surface-mounted sensor experiences a localized tensile strain (ϵ) that is geometrically proportional to the curvature and the distance from the neutral axis of the substrate. By correlating the sensor's resistance variations ($\Delta R/R_0$) recorded during the uniform loop tests with the quasi-static stretching data, a direct analytical transfer function between the real-time electrical readout and the bending curvature (κ) was established.

C. Impedance Measurement and ADC Conversion

The signal acquisition circuitry relies on a precision voltage divider network to measure the unknown tissue impedance (Z_x). The reference resistor was deliberately selected as $R_{ref} = 2.2 \text{ k}\Omega$. This specific value was chosen because the maximum sensitivity and accuracy of a voltage divider occur when the known reference resistance closely matches the target unknown resistance. The baseline impedances of blood, vessel walls, and clot materials all fall near this 2.2 k Ω threshold; as measured values deviate further from this reference point, the mathematical conversion becomes increasingly non-linear and less accurate.

The analog voltage drop across the sensor is captured by a microcontroller equipped with a 10-bit Analog-to-Digital Converter (ADC). The fundamental voltage divider equation governs the relationship between the applied voltage (V_{in}) and the measured voltage drop (V_{out}):

$$V_{out} = V_{in} \left(\frac{Z_x}{Z_x + R_{ref}} \right)$$

Algebraically rearranging this formula to solve for the unknown tissue impedance yields the standard analog voltage equation:

$$Z_x = R_{ref} \left(\frac{V_{out}}{V_{in} - V_{out}} \right)$$

To process this in the digital domain, the ADC translates the continuous voltage into a discrete binary value (D) ranging from 0 to 1023. By substituting the ratio of the discrete ADC reading to the maximum 10-bit resolution ($D/1023$) for the analog voltage ratio (V_{out}/V_{in}), the system calculates the localized tissue impedance using the final digital conversion formula:

$$Z_x = R_{ref} \left(\frac{D}{1023 - D} \right)$$

D. Differential Impedance Analysis (ΔZ)

A major algorithmic consideration in this design is that the absolute resistance values of the target biological materials are exceptionally close to one another. Furthermore, when the probe is submerged in blood or water, the surrounding fluid creates a parallel conductive circuit. This parallel path systematically depresses the calculated absolute impedance and can vary dynamically over time as fluid flow and composition change.

To account for this, the detection logic does not rely on static baseline measurements. Instead, the system evaluates the target material by tracking the dynamic changes in impedance

(ΔZ) detected by the instrument. By calculating the relative impedance shifts as the probe navigates the environment, the system effectively isolates the target tissue signatures from the variable parallel fluid resistance.

E. Hysteresis Signal Processing

A critical issue affecting carbon nanotube-polymer composite sensors is their inherently pronounced hysteresis behavior. Hysteresis arises from the viscoelastic response of the polymeric substrate, such as PI or PDMS, and the microscopic sliding or rearrangement of the CNT network during loading and unloading. As a result, the measured resistance during the loading phase may differ from the resistance during the unloading phase even under the same deformation state. This path-dependent response can reduce the accuracy of dynamic strain and curvature tracking.

In this preliminary study, hysteresis was treated as an important limitation rather than a fully solved problem. Classical operator-based models, such as the Prandtl-Ishlinskii (P-I) model and the Preisach model, were considered as possible compensation frameworks for future implementation. These models are commonly used to describe memory-dependent hysteresis loops and could be used to estimate an inverse compensation function if sufficient cyclic loading data are available.

However, accurate hysteresis compensation requires repeated loading and unloading experiments under different amplitudes and deformation rates. Since the current work mainly focused on basic tensile calibration and curvature-response validation, a complete inverse feedforward compensation model was not yet implemented. Future work will collect cyclic deformation datasets to identify hysteresis model parameters and evaluate whether inverse compensation can improve real-time curvature estimation.

V. EXPERIMENTAL SETUP

A. Human Skin and Tissue Phantom Fabrication

A heterogeneous, multi-layered tissue phantom of the femoral region was fabricated using varying formulations of Eco-flex silicone. To prepare the simulated vasculature, gelatin was injected into a smaller tube representing the femoral artery, and both this artery and a larger tube representing the femoral vein were sealed to prevent silicone ingress. The phantom was molded sequentially to mimic distinct layers of biological tissue. All silicone formulations were mixed at a 1:1 ratio of Part A to Part B to ensure proper curing.

- **Deep Fascia (Muscle):** A base layer consisting of 25 mL of Ecoflex 00-45 was poured. This cured for 40 minutes until it was tacky, at which point the simulated femoral vein and artery were positioned side-by-side. These vessels were placed approximately 1.7 inches from the final external surface [6].
- **Subcutaneous Fat:** A 350 mL volume of Ecoflex 00-10 was poured over the vessels and muscle layer. This layer cured for an additional 40 minutes until tacky but not completely dry.

- **Dermis and Epidermis:** A mesh material was embedded on top of the fatty layer to simulate the structural grit of the upper tissue. Finally, 30 mL of Ecoflex 00-30 was poured over the mesh to seal the top and complete the phantom structure.

B. Curvature Sensing Setup via Circular Loop Method

To establish the relationship between the sensor's electrical output and macroscopic structural bending, a circular-loop-based curvature sensing setup was designed. Flexible polyurethane tubes of varying predefined lengths (L) can be used as support substrates. By joining the ends of each tube to form a closed circular loop, an approximately uniform bending radius can be generated, where $r = L/(2\pi)$. Consequently, the applied curvature can be estimated as $\kappa = 1/r = 2\pi/L$.

In this setup, the flexible CNT sensor is conformally attached to the outer surface of the tube. When the tube is bent into a loop, the outer surface experiences tensile strain, causing a corresponding change in the CNT sensor resistance. By recording the steady-state resistance values under different loop lengths, an empirical relationship between sensor resistance and bending curvature can be obtained. This setup provides a practical method for future curvature calibration and shape-sensing validation.

C. Tissue Phantom Penetration and Proximal Force Profiling

To evaluate the response of the sensing system under a clinically relevant puncture scenario, a tissue phantom penetration experiment was conducted using a standard medical syringe needle. The tissue phantom was designed as a simplified multi-layer structure to mimic the mechanical heterogeneity of human skin, including layers analogous to the epidermis, dermis, and subcutaneous fat. This layered structure provides distinct mechanical interfaces, allowing the insertion process to generate measurable changes in reaction force as the needle passes through different material boundaries.

During the experiment, the syringe needle was inserted vertically into the tissue phantom. A commercial thin-film piezoresistive pressure sensor was attached to the thumb-rest area of the syringe plunger, which corresponds to the location where force is applied by the operator. This configuration was used to measure proximal force feedback, representing the mechanical resistance transmitted from the needle-phantom interaction through the syringe body to the operator's finger. Although this measurement is indirect, it provides a practical baseline for evaluating the force variation during puncture.

The expected force response during insertion includes a gradual increase in force as the needle compresses the phantom surface, followed by a local peak when the needle tip punctures a material layer. After rupture, the force may drop suddenly as the needle enters the next layer. In a multi-layer phantom, this buildup-peak-drop pattern can occur multiple times, corresponding to the needle crossing different tissue-like interfaces. Therefore, the proximal force profile can be used to identify potential puncture events and compare the mechanical differences among phantom layers.

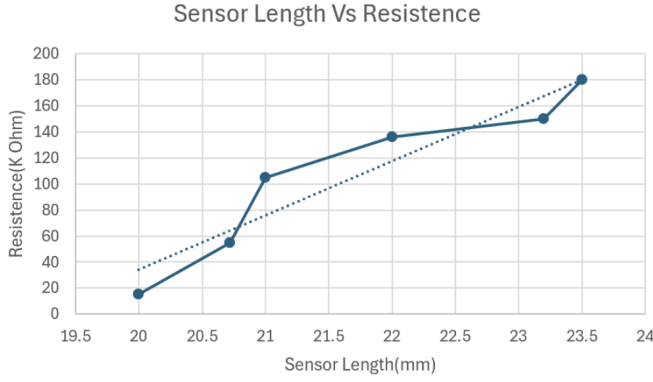


Fig. 1. Stretch vs. resistance response of the CNT sensor

This experiment provides a baseline for conventional proximal sensing and helps motivate future distal sensing approaches. Since the pressure sensor is mounted on the syringe plunger rather than directly at the needle tip, the measured signal may include additional effects such as syringe friction, hand motion, and non-axial loading. Future work can compare this proximal force signal with flexible CNT or FPCB sensor measurements placed closer to the needle–tissue interface to evaluate whether localized sensing improves boundary detection accuracy.

D. Underwater Prototype Validation

To validate the parallel circuit compensation logic and the waterproof integrity of the sensor design, the handmade single-channel prototypes were subjected to submerged underwater testing. These trials successfully demonstrated the sensing methodology’s viability while operating in a highly conductive fluid environment. The underwater data confirmed the theoretical decrease in overall impedance caused by the parallel fluid circuit, ultimately validating the necessity of using the differential ΔZ approach to reliably detect material boundaries during active fluid immersion.

VI. EXPERIMENTAL RESULTS AND ANALYSIS

A. Quasi-Static Tensile Calibration and Piezoresistive Response

The fundamental electromechanical response of the CNT sensor was first evaluated under quasi-static tensile loading. The sensor’s absolute resistance was recorded as its length was incrementally stretched from an initial baseline of 20.0 mm up to 23.5 mm.

The experimental data demonstrates a strong positive correlation between the applied tensile strain and the measured macroscopic resistance, confirming the underlying piezoresistive mechanism of the CNT network. Specifically, the resistance escalated from approximately 15 k Ω at the resting state to nearly 180 k Ω at maximum elongation. However, it is crucial to note that the resistance-strain relationship exhibits distinct non-linear characteristics. While a linear regression (dashed line) provides a general trend, the actual sensor response (solid line) displays varying sensitivities (Gauge

Factors) across different strain regions. For instance, a highly sensitive transition occurs between 20.7 mm and 21.0 mm, followed by a region of reduced sensitivity. This inherent non-linearity highlights the insufficiency of simple linear scaling and experimentally validates the necessity for the advanced, non-linear calibration mapping and hysteresis compensation frameworks proposed in this study.

B. CNT Sensor Performance and Compensation Results

VII. DISCUSSION

The preliminary experimental results presented in this study validate the basic feasibility of utilizing flexible CNT networks for spatial tracking and tissue analysis, while simultaneously exposing several critical metrological limitations that must be addressed in future iterations.

A. Non-linearity and the Hysteresis Bottleneck

While the quasi-static tensile calibration demonstrated a clear piezoresistive response, the relationship between strain and resistance is highly non-linear. The varying gauge factor across different elongation states complicates simple mathematical mapping. Furthermore, polymeric nanomaterial composites inherently suffer from pronounced hysteresis due to the viscoelastic relaxation of the substrate and the internal friction of the CNT network during unloading. Although an algorithmic hysteresis compensation framework was proposed in Section IV, fully realizing its potential requires comprehensive cyclic loading datasets (loading and unloading phases) that were outside the scope of this preliminary phase. Without an accurate inverse feedforward model, real-time spatial tracking will suffer from path-dependent memory effects, severely degrading measurement fidelity during dynamic surgical maneuvers.

B. Proximal versus Distal Sensing Paradigms

A primary motivation of this project is to transition from subjective proximal haptic feedback to objective distal localized sensing. In the current phantom penetration setup, the piezoresistive force sensor was mounted on the proximal plunger (the finger interface). While this successfully registered the macroscopic force variations required to puncture distinct phantom layers (epidermis, dermis, and subcutaneous fat), the acquired data still represents an indirect measurement. Proximal signals are inherently corrupted by mechanical noise, such as the sliding friction of the syringe barrel and non-axial human tremors. To achieve true direct localized sensing, future iterations must integrate ultra-thin FPCB or CNT arrays directly onto the distal shaft of the insertion instrument. Comparing the current proximal baseline data with future distal sensor data will be crucial to quantify the reduction in mechanical noise and the improvement in tissue boundary detection accuracy.

C. Dynamic Response Constraints

Finally, clinical puncture procedures are highly dynamic. The current quasi-static calibration does not account for the sensor's temporal step-response at varying insertion velocities. Future evaluations must assess the dynamic bandwidth of the sensors to ensure that high-frequency pressure drops—such as the instantaneous rupture of a delicate tissue membrane—are captured without algorithmic lag or hardware attenuation.

VIII. CONCLUSION

This paper presented a foundational investigation into the application of flexible piezoresistive CNT and FPCB sensors for dynamic shape sensing and biomechanical tissue puncture analysis. To evaluate the sensor performance in a clinically relevant context, custom multi-layer tissue phantoms replicating human skin biomechanics were formulated and subjected to controlled syringe penetrations. The preliminary findings confirm the fundamental piezoresistive mapping capabilities of the sensors and successfully established a proximal force profile baseline during structural tissue puncturing.

Additionally, the successful underwater validation of the handmade single channel prototypes demonstrated the viability of localized electrical tissue differentiation. By utilizing a 2 ms pulsed DC excitation strategy and analyzing relative impedance shifts (ΔZ), the prototype successfully isolated target material signatures from the parasitic parallel resistance of surrounding fluids. This differential logic proves essential for maintaining accuracy during active fluid immersion, paving the way for the future fabrication and deployment of the designed multi-channel FPCB.

However, as a preliminary proof-of-concept, this study identified significant hardware and signal processing constraints. The inherent non-linear sensitivity and anticipated viscoelastic hysteresis of the CNT sensors demand the implementation of robust mathematical compensation models, such as the Prandtl-Ishlinskii framework, trained on comprehensive cyclic loading data. Furthermore, the current reliance on proximal force measurement highlights the necessity for direct distal integration. Future work will explicitly focus on embedding flexible sensor arrays directly at the distal needle-tissue interface to eliminate proximal mechanical noise. By optimizing the hardware integration and refining the real-time hysteresis algorithms, this flexible sensing paradigm holds substantial promise for advancing autonomous soft robotics, objective surgical skill evaluation, and real-time medical diagnostics.

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